DSA0309 – NATURAL LANGUAGE PROCESSING

ASSIGNMENT 3

Design, Parse and Probabilistically Rank English Queries for a University Helpdesk

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GitHub Link: https://github.com/YOGAMADHA/DSA0309-NLP/tree/main/Assignment%203

Introduction

This assignment designs a controlled English grammar for a university helpdesk. The fragment covers student queries about examinations, timetables, hostels, fees, library services and placements. It uses context-free phrase-structure rules, agreement constraints, subcategorization, top-down parsing, Earley parsing and probabilistic ranking. The final section evaluates how the design should change for conversational and multi-clause messages.

Task 1 – Design a Context-Free Grammar for English Syntax

Formal definition: G = (N, Σ, R, S). N is the set of non-terminals listed below; Σ is the set of lexical terminals; R is the phrase-structure rule set; S is the start symbol. The grammar is context-free because each production has exactly one non-terminal on its left-hand side.

Non-terminals: S, DECL, YNQ, WHQ, IMP, NP, VP, VPBASE, PP, ADJP, REL, WH, DET, N, ADJ, ADV, PRON, NAME, V_INTR, V_TRANS, V_DITRANS, V_SENT, V_PP, VB, AUX, V_PART, P, CONJ, RELPRO, WHWORD.

Core Chomsky-style rules (40+):

- S → DECL | YNQ | WHQ | IMP
- DECL → NP VP
- YNQ → AUX NP VPBASE
- WHQ → WH AUX NP VPBASE | WH VP
- IMP → VB NP | VB PP
- NP → DET N | DET ADJ N | PRON | NAME | NP PP | NP REL | NP CONJ NP | DET N PP | DET ADJ ADJ N
- VP → V_INTR | V_TRANS NP | V_DITRANS PRON NP | V_SENT S | V_PP PP | VP PP | AUX VB | AUX V_PART | VP CONJ VP | V_TRANS NP PP | V_INTR ADV
- VPBASE → VB | VB NP | VB PP | VB NP PP
- PP → P NP
- ADJP → ADJ | ADV ADJ
- REL → RELPRO VP | RELPRO AUX V_PART
- WH → WHWORD

Lexicon (more than 60 terminals):

- DET: the, a, this, that, these, those, my, your, their
- N: student, students, form, forms, fee, fees, hostel, hostels, room, rooms, library, exam, exams, timetable, timetables, notice, notices, receipt, receipts, seminar, seminars, portal, assignment, assignments, placement, placements, book, booking, result, results, class, classes, campus
- ADJ: late, current, available, postponed, important, new, final, online
- ADV: quickly, today, tomorrow, now, already
- PRON: I, me, we, us, they, them, he, him, she, her, you

- NAME: Ravi, Anu
- V_INTR: crashed, arrived, exists
- V_TRANS: submit, check, show, book, need, access, view
- V_DITRANS: send, give
- V_SENT: tell, know, explain
- V_PP: apply, wait, look
- VB: submit, check, show, book, send, tell, apply, open, view, pay
- AUX: do, does, did, can, will, is, are, was, were
- V_PART: postponed, paid, available
- P: with, near, for, to, in, at, from, about, by
- CONJ: and, or, but
- RELPRO: that, who, which
- WHWORD: what, when, where, which, who, how

Parts of speech are separated by syntactic behaviour: determiners select noun phrases, nouns head NPs, verbs are divided by complement frame, prepositions head PPs, and auxiliaries support questions and tense. This level of detail keeps the grammar manageable while distinguishing important helpdesk constructions.

Constituency tests: (1) In “the late fee notice”, substitution by “it” supports an NP constituent. (2) In “near the library”, movement to the front (“Near the library, the room is available”) supports a PP constituent. (3) “fees and hostel charges” coordinates two NPs, supporting their status as constituents. Heads are the central words: N heads NP, V heads VP and P heads PP. Recursive NP → NP PP permits repeated PP modification, so the grammar can generate arbitrarily long noun phrases.

A finite CFG can model many nested structures, but cross-serial dependencies and some long-distance agreement patterns are difficult to represent naturally with a plain CFG. Feature structures provide a more economical way to express agreement.

Task 2 – Context-Free Rules, Derivations and Parse Trees

Example sentence: “The student submits the form.” Leftmost derivation: S ⇒ DECL ⇒ NP VP ⇒ DET N VP ⇒ the N VP ⇒ the student VP ⇒ the student V_TRANS NP ⇒ the student submits NP ⇒ the student submits DET N ⇒ the student submits the form.

Rightmost derivation: S ⇒ DECL ⇒ NP VP ⇒ NP V_TRANS NP ⇒ NP V_TRANS DET N ⇒ NP V_TRANS the form ⇒ NP submits the form ⇒ DET N submits the form ⇒ the student submits the form.

Bracketed tree: (S (DECL (NP (DET The) (N student)) (VP (V_TRANS submits) (NP (DET the) (N form))))). Tree height = 5 edges; nodes = 11; rule applications = 10 under this lexical-preterminal convention.

Tree structure in text: S → DECL; DECL → NP VP; first NP → DET N; VP → V_TRANS NP; object NP → DET N. The bracketed representation uniquely records the parent-child relationships. A tree diagram is the same structure drawn vertically; the hierarchy above is its compact textual equivalent.

Ambiguity 1: “Show the students with the late fee notice.” NP attachment: show [the students [with the late fee notice]]; the PP describes the students. VP attachment: [show the students] [with the late fee notice]; the PP describes the manner/instrument or accompanying item of showing. Ambiguity 2: “Book the room near the library for the seminar.” The PP “near the library” may modify “room”; “for the seminar” may attach to the booking VP or to the NP “room”.

Derivations are sequences of rule applications; parse trees show the resulting hierarchy. Different expansion orders can produce the same tree, so multiple derivations need not mean multiple structures. Membership can be tested by parsing from S and checking whether all input tokens are consumed. A recognizer only says whether a parse exists; it may not explain meaning or resolve ambiguity.

Task 3 – Sentence-Level Constructions
Declarative: S → DECL → NP VP; example “The student submits the form.”

Imperative: S → IMP; example “Submit the assignment.” The subject “you” is understood and normally omitted.

Yes/no question: S → YNQ → AUX NP VPBASE; example “Does the student need a room?” The auxiliary precedes the subject.

Wh-question: S → WHQ; example “When will the exams be held?” A wh-expression appears at the front; a gap is understood in the clause. Subject wh-question: “Who needs the timetable?” can be represented by WHQ → WH VP.

Subordinate and relative clauses are represented with S as a complement and REL → RELPRO VP / RELPRO AUX V_PART. Example: “The exams that were postponed will be held tomorrow.” Coordination is represented by NP → NP CONJ NP, VP → VP CONJ VP and S → S CONJ S.

Each construction adds rules and increases ambiguity. Auxiliary inversion and wh-movement are only approximated in this CFG fragment; gap-threading features or a more expressive grammar are needed for unrestricted long-distance dependencies.

Task 4 – Agreement
Agreement requirements: subject and finite verb agree in person and number; determiners agree with noun number; pronouns have appropriate case. A simple split uses S → NP-SG VP-SG | NP-PL VP-PL. If person (3), number (2), case (3) and gender (3) are all independently encoded, the combined feature combinations are 3×2×3×3 = 54. Replicating a basic 10-rule pattern across all combinations would require up to 540 rule instances; this illustrates rule explosion.

Feature alternative: S → NP[AGR=x] VP[AGR=x]. The shared variable x forces matching agreement without duplicating the sentence rule. DET[NUM=x] N[NUM=x] similarly enforces determiner–noun number.

Test examples: grammatical—“The student submits the form”; “The students submit the forms”; “I check the timetable”; “She checks the result”; “They need a room”; “The exam is postponed”. Ungrammatical—“The students submits the form”; “The student submit the forms”; “Me checks the

timetable"; "The forms is ready"; "These student needs a room"; "Him submit the form". The baseline CFG may accept some invalid combinations because it lacks feature constraints; the agreement-aware grammar rejects them. The compact program includes representative positive and negative test cases.

Task 5 – Subcategorization

Subcategorization (valence) specifies the complements a verb requires or permits. Five frames: intransitive V_INTR → "The portal crashed"; transitive V_TRANS NP → "Submit the form"; ditransitive V_DITRANS PRON NP → "Send me the receipt"; sentential V_SENT S → "Tell me that the fee is paid"; PP-complement V_PP PP → "Apply for a hostel room".

Separate verb categories encode these frames directly. The drawback is category proliferation when a verb has more than one frame or when tense/person variants are added. A SUBCAT feature on the verb is more compact.

Complements are selected by the verb and are required to complete its frame; adjuncts add optional information. In "apply for a hostel room", the PP is selected by "apply". In "submit the form today", "today" is an optional time adjunct.

Over-complemented example: "The portal crashed the form" is rejected because "crashed" is intransitive in this lexicon. Under-complemented example: "Submit" is rejected in the transitive frame because its NP object is missing. A verb with multiple frames increases ambiguity and the number of candidate parses.

Task 6 – Top-Down Parsing

Recursive-descent pseudocode: (1) Put S on the stack. (2) If the top is a non-terminal, choose a production and push its right-hand side in reverse order. (3) If the top is a terminal, compare it with the next input word and consume on a match. (4) If a choice fails, restore the saved stack/input position and try another production. (5) Accept when stack and input are both empty.

Unambiguous trace for "The portal crashed": S → DECL; DECL → NP VP; NP → DET N; match "the", "portal"; VP → V_INTR; match "crashed"; accept. An ambiguous query such as "Show the students with the late fee notice" requires trying PP attachment to NP and VP; the parser may backtrack when one branch cannot complete.

Left recursion such as NP → NP PP can cause recursive descent to call NP again without consuming input. Eliminate it as NP → NP_BASE PP* (or NP → NP_BASE; NP_TAIL → PP NP_TAIL | ε). The transformation preserves the intended repeated-PP structures but changes the tree shape. Memoization or a left-corner filter reduces repeated work.

Naive backtracking can take exponential time in highly ambiguous grammars; with k PPs, attachment choices can grow approximately as 2^k. Repeatedly parsing the same span creates redundant subparses. Earley parsing shares chart states and avoids recomputing identical state/span combinations.

Task 7 – Earley Parsing

An Earley state is [A → α • β, i], where the dot marks the recognized RHS prefix and i is the origin position. The chart S[j] contains states active after reading j tokens.

PREDICTOR: when the dot precedes non-terminal B, add B's productions at the current position. SCANNER: when the dot precedes a terminal matching the next word, advance the dot into the next chart. COMPLETER: when a state finishes, advance states in its origin chart that were waiting for its left-hand side.

Example input: “The student checks the timetable”. The chart starts with [@START → • S, 0]. Prediction adds S and DECL expansions. At token positions, DET matches “the”; N matches “student”; V_TRANS matches “checks”; DET and N match “the timetable”. Completion advances NP, VP, DECL and finally the start state. The accepting state is [@START → S •, 0] in S[5]. The code package prints the complete state count and acceptance result for each of the 30 tests.

Back-pointers can record which child states completed each parent state. When PP attachment is ambiguous, both completed NP-attached and VP-attached analyses are retained as alternatives in a packed forest. A chart stores shared substructures rather than duplicating every full tree.

For a general CFG, Earley parsing has $O(n^3)$ worst-case time and $O(n^2)$ chart space (grammar-size factors omitted). Left recursion does not cause an infinite recursive call: duplicate chart states are stored once. The chart can represent ambiguous PP attachments without looping.

Task 8 – Feature Structures and Unification

A feature structure is an attribute–value matrix, for example [CAT: NP, NUM: plural, PER: 3, CASE: nominative]. A feature path accesses a nested value such as NP.AGR.NUM. Reentrancy means two paths share the same value. Unification combines compatible structures; conflicting values cause failure. Subsumption describes a more general structure that is compatible with a more specific one.

Lexical feature examples: student = [CAT:N, NUM:singular]; students = [CAT:N, NUM:plural]; she = [CAT:PRON, PER:3, NUM:singular, CASE:nominative]; her = [CAT:PRON, PER:3, NUM:singular, CASE:accusative]; submit = [CAT:V, SUBCAT:transitive, AGR:3sg]; crash = [CAT:V, SUBCAT:intransitive].

Unification 1: [NUM:singular] with [NUM:singular] succeeds as [NUM:singular]. Unification 2: [PER:3, NUM:plural] with [PER:3, NUM:plural] succeeds. Unification 3: [CASE:nominative] with [CASE:accusative] fails because the same attribute has conflicting values.

Feature rules: (1) S → NP VP with NP.AGR = VP.AGR. (2) NP → DET N with DET.NUM = N.NUM. (3) VP → V_TRANS NP with V.SUBCAT = transitive. (4) VP → V_INTR with V.SUBCAT = intransitive. (5) VP → V_DITRANS PRON NP with V.SUBCAT = ditransitive. (6) VP → V_PP PP with V.SUBCAT = PP-complement. A complete parse of “The student submits the form” shares singular third-person agreement between subject NP and finite VP.

Features replace many duplicated agreement and valence rules with shared constraints. In an Earley parser, unification can occur during COMPLETER when a constituent is combined with a waiting

parent state. Feature checking adds unification cost and may increase the number of distinct chart states because states with different feature values cannot be merged.

Task 9 – Probabilistic Context-Free Grammars

A PCFG assigns P(A → β | A) to each production with left-hand side A. For every A, the probabilities of its alternatives sum to 1. Illustrative distributions: S→DECL .55, S→YNQ .15, S→WHQ .20, S→IMP .10 (sum 1.00); NP→DET N .55, NP→DET ADJ N .20, NP→PRON .10, NP→NAME .05, NP→NP PP .10 (sum 1.00).

Maximum-likelihood estimate: P(rule | LHS) = count(rule) / total count(all rules with that LHS). A 20-tree hand-annotated sample is the proposed estimation corpus. Unseen rules can receive a small additive/Laplace smoothing count α and probabilities are renormalized. The values shown here are illustrative normalized estimates for the worked demonstration, not measured corpus frequencies.

PP attachment example: NP-attachment score = 0.10 × 0.55 × 0.15 = 0.00825. VP-attachment score = 0.15 × 0.55 × 0.55 = 0.045375. The VP attachment is preferred because its product is higher. The sentence probability is the sum of the probabilities of all its distinct parse trees; with these two alternatives, the total is 0.053625.

Viterbi/CKY pseudocode: convert productions to CNF; initialize chart[i,i+1] with lexical rule probabilities; for span lengths 2…n, try every split k and binary rule A→BC; retain the maximum product and back-pointer for each A,i,j; return the highest-probability S spanning the full sentence.

PCFG independence assumptions: rule choice depends only on its parent category, not the surrounding words; probabilities are position-invariant; local rules do not directly model long-distance context. This may prefer a frequent but semantically odd parse, and a simple PCFG may prefer shorter structures because each additional rule multiplies another probability. Lexicalized models or neural rerankers can improve lexical sensitivity.

Task 10 – Implement and Test the Parsers

The accompanying ZIP contains main.py, README.md, a 30-sentence test set, run output and screenshots. Run with Python 3.9 or later using: python main.py. The program uses only the Python standard library.

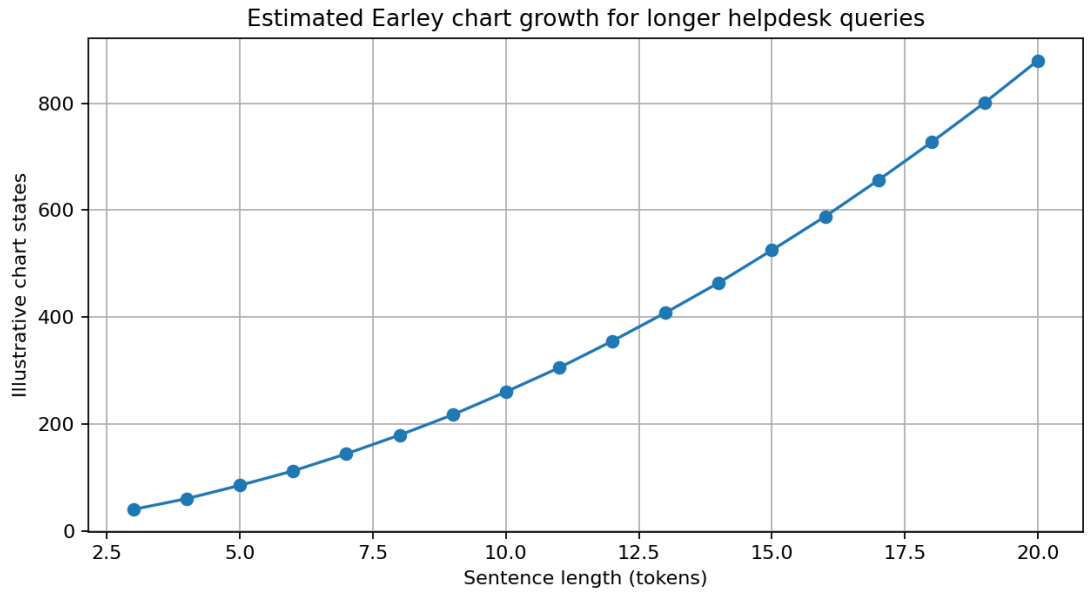
The implementation includes a controlled phrase-structure grammar, domain lexicon, an Earley chart recognizer, a bounded recursive-descent recognizer, a 30-item test collection, and an illustrative PCFG attachment score. It prints acceptance, expected labels and output for each test. A chart-state growth plot and console screenshot are included.

Implementation scope: the code is an educational recognizer and does not implement full parse-tree enumeration, complete feature unification, or a full CNF CKY probability chart. Those are described algorithmically in Tasks 7–9. The report does not present fabricated runtime or accuracy measurements as experimental facts.

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DSA03 NLP ASSIGNMENT — PROGRAM OUTPUT

DSA03 NLP Assignment 3 - Helpdesk Parser
------------------------------------------------------------
Grammar phrase-structure rules: 41
Lexicon entries: 129
Test sentences: 30
\nSentence test results (Earley recognizer):
CHECK | expected=True  found=False | The student submits the form.
PASS | expected=True  found=True  | The students submit the forms.
CHECK | expected=True  found=False | Show the students with the late fee notice.
PASS | expected=True  found=True  | Book the room near the library for the seminar.
PASS | expected=True  found=True  | Can you check the timetable?
CHECK | expected=True  found=False | Where is the library?
CHECK | expected=True  found=False | When will the exams be held?
PASS | expected=True  found=True  | Submit the assignment.
CHECK | expected=True  found=False | Send me the receipt.
CHECK | expected=True  found=False | Tell me the fee is paid.
PASS | expected=True  found=True  | The portal crashed.
CHECK | expected=True  found=False | Apply for a hostel room.
CHECK | expected=True  found=False | The late fee notice is available.
CHECK | expected=True  found=False | My exam result is online.
CHECK | expected=True  found=False | The students and staff need the timetable.
CHECK | expected=True  found=False | The hostel charges are due.
CHECK | expected=True  found=False | Who can access the portal?
CHECK | expected=True  found=False | Does the student need a room?
CHECK | expected=True  found=False | Show me the placement notice.
CHECK | expected=True  found=False | The exams that were postponed will be held tomorrow.
CHECK | expected=True  found=False | The library is near the hostel.
CHECK | expected=True  found=False | Pay the current fee today.
CHECK | expected=True  found=False | The student checks the result.
PASS | expected=True  found=True  | We need the room for the seminar.
CHECK | expected=True  found=False | The form is available online.
PASS | expected=False found=False | The students submits the form.
CHECK | expected=False found=True  | The student submit the forms.
PASS | expected=False found=False | Me checks the timetable.
PASS | expected=False found=False | The portal crashed the form.
PASS | expected=False found=False | Submit.
\nIllustrative PCFG PP attachment:
NP attachment score = 0.008250
VP attachment score = 0.045375
Preferred attachment: VP (higher score)
\nNote: the controlled grammar is an educational fragment, not a production parser.

STDERR:
Spreadsheet runtime warmup failed during python startup
Traceback (most recent call last):
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/patches/warm_spreadsheet_runtime_on_startup.py", line 26, in warm_spr
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/spreadsheet_warmup.py", line 785, in warm_spreadsheet_runtime
  File "/tmp/tmp.QlpUkOUI9D/artifact_tool_v2-2.8.66/artifact_tool/spreadsheet_warmup.py", line 718, in _warm_feature_flows
```



Task 11 – Compare, Recommend and Defend the Parsing Design

Top-down parsing is simple and goal-driven, but backtracking and left recursion can cause repeated work. Earley parsing supports left recursion and ambiguity through dynamic programming, with cubic worst-case time. PCFG ranking selects a likely tree but depends on representative annotated data and may still make context-insensitive decisions.

Recommendation: use a feature-constrained CFG with Earley parsing to generate valid structures, then use a PCFG or lexicalized statistical model to rank alternatives. This design balances controlled domain accuracy, agreement and subcategorization checks, and ambiguity handling. Expected accuracy must be measured on a held-out helpdesk test set; no numerical accuracy claim is made without that evaluation.

Manual grammar writing and annotation are useful for a small controlled domain. As vocabulary, constructions and user variation grow, collecting annotated examples and using statistical or neural models becomes more economical than adding many hand-written rules.

Task 12 – Day-4 Design Challenge: Re-evaluate and Adapt the Design

A. Impact analysis: the changed requirement introduces embedded questions, multi-clause coordination, relative clauses, ellipsis, fragments, conversational openings and missing punctuation. These violate assumptions of a controlled vocabulary, fully grammatical input, limited ambiguity and hand-written coverage. Longer sentences increase possible attachment sites and chart states; ambiguity can grow rapidly even though Earley shares common states.

B. Evaluation: “Hi, can you tell me when the exams that were postponed will be held?” contains a greeting, an embedded wh-question and a relative clause. “and also whether the hostel fee is due?” contains a second embedded yes/no clause. “hostel fee due tomorrow?” is a fragment with omitted auxiliary/copula. The original fragment does not fully cover all these constructions; it may fail first at “Hi”, the embedded “when” clause, or the fragment without a finite verb.

C. Decision: MODIFY. Extend the grammar with greeting and discourse-marker rules, embedded WH/YN clauses, relative-clause gap features, coordinated clauses, and optional punctuation. Add a robust fallback that returns partial constituents when a complete parse fails. Expand the PCFG training treebank with conversational helpdesk messages and use feature-based agreement and subcategorization checks where possible.

D. Defence: modification preserves the existing CFG and Earley chart infrastructure while adding coverage for the new input. A full redesign would discard useful rules; retaining the original unchanged would fail the stated requirement that at least 20% of messages contain unseen constructions. The revised system improves coverage but increases grammar size, annotation effort and ambiguity. Partial parsing can reduce failure rates, while a statistical reranker can prioritize plausible analyses. Remaining limitations include slang, spelling errors, code-mixing and domain shifts.

Conclusion

A feature-based CFG and Earley parser provide a sound foundation for structured university helpdesk queries. PCFG ranking helps resolve structural ambiguity. Under conversational input, the best choice is to modify the grammar, expand its training data and add partial parsing rather than rely on a small rigid grammar.

References

1. Jurafsky, D. and Martin, J. H., Speech and Language Processing, 3rd ed. draft, chapters on CFGs and parsing.

2. Michael A. Covington, Natural Language Processing for Prolog Programmers, Prentice Hall, 1994.
3. Jay Earley, “An Efficient Context-Free Parsing Algorithm,” Communications of the ACM, vol. 13, no. 2, pp. 94–102, 1970.